

THURA ZAW

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First Class Honors graduate in **Robotics & AI Engineering** from KMITL with **9 months** of industry experience spanning embedded firmware, PCB design, ROS2, and NVIDIA Isaac Sim. Built real-world systems including a 12-DOF humanoid robot leg and an autonomous UAV-UGV inspection platform. Seeking Robotics, Embedded Systems, or Mechatronics Engineering roles in **Bangkok**.

WORK EXPERIENCE

General Magic (Thailand) Co., Ltd

Jun. 2025 – Mar. 2026

Robotics System Engineer

Bangkok, TH

- Designed and implemented the electrical system for a bipedal humanoid robot leg, enabling low-latency communication across **12 actuator joints** using **CAN Bus** protocol and **SocketCAN** on **ROS2 control** framework
- Developed **2 custom PCBs** and corresponding embedded firmware for actuator control utilizing tools such as KiCAD, LTSpice and STM32 Cube IDE
- Collaborated within a team of 4 to achieve both **sim-to-sim** and **sim-to-real** prototyping in **NVIDIA Isaac Sim & MuJoCo**, with **50 Hz real-time control loop** and **200 Hz** for **IMU data reading** and **actuator's position and velocity commands**.
- Coordinated efforts between **KMITL** capstone requirements and company product goals, contributing technical expertise to a **joint academic-industry** robotics initiative

National Institute of Technology (Sendai College)

May 2025 – Jun. 2025

Robotics System Engineer Intern

Sendai, JP

- Created an **attention robot** prototype using C++, Arduino, servo control, and OLED display, enabling **real-time danger alerts** for technicians through expressive actions and sound

EDUCATION

King Mongkut's Institute of Technology Ladkrabang (KMITL)

Apr. 2026

Bachelor of Engineering in Robotics and Artificial Intelligence Engineering

Bangkok, TH

- GPA: 3.74 / 4.0; First Class Honors**
- Relevant Coursework:** Mobile Robots, Introduction to Robotics & AI, Feedback Control, **Kinematics & Dynamics**, Computer Vision, Artificial Intelligence Technology, **Microprocessor & Interface**, Introduction to Algorithms, Differential Equations & Matrix Algebra, Probability & Statistics, **Reinforcement Learning**

AWARDS & RECOGNITION

K-Engineering World Tour 2024's Most Innovative Project Award

- Designed a two-dimensional **Cartesian Robot** for drawing photorealistic images

KMITL's Line Following Robot 2nd Prize Winner

- Designed a **STM32** high speed line following robot using **Nucleo STM32F411RE**, IR sensors and PID control

PROJECTS (Selected Projects — full case studies at [Portfolio](#))

Transformable Hybrid UAV-UGV System for Visual Inspection in Unstructured Environment

- Led design and implementation of a morphing multi-joint UAV-UGV platform using **ROS2 Humble & PX4 Autopilot** for post-earthquake search and rescue in **GPS-denied** environments
- Developed custom-trained **crack detection** and human detection AI models using **Roboflow** for real-time survivor identification among debris.

Bipedal Humanoid Robot Leg with Natural Walking Gait

- Led electrical system design and embedded hardware development for a **CANBus Splicer Power Delivery PCB** (120 ohm controlled impedance) for signal and power integrity across **12 CubeMars** actuators using **1 Mbps ISO CANBus** best practices.
- Development of **droop-based current balancing** solution for combining 6 channels of MeanWell power supplies into a single **48V 120A** robot's power station using **ideal diode controller IC** and high power **MOSFETs**.

VOLUNTEER EXPERIENCE

King Mongkut's Institute of Technology Ladkrabang (KMITL)

Jul. 2024 – Mar. 2025

Teaching Assistant (TA)

Bangkok, TH

- Assisted in designing and constructing laboratory experiments and demonstrations to enhance student understanding of complex concepts.
- Provided technical assistance and troubleshooting support for students during laboratory sessions, ensuring safe and effective use of electrical equipment.

CERTIFICATIONS, SKILLS, TECHNOLOGIES, LANGUAGES & INTERESTS

- Certifications:** Altium Education PCB Design Course; Google Data Analytics (Coursera); Introduction to Linux (LFS101) (The Linux Foundation); Certified Jetson AI Competency User (**NVIDIA**)
- Skills:**
 - Robotics & Software:** ROS2, C++, Python, SLAM, Computer Vision, Deep Learning, Sensor Fusion, Control Systems
 - Hardware & Electronics:** PCB Design, Embedded Systems, CAN Bus, STM32, KiCAD, LTSpice, Altium Designer
 - Tools & Platforms:** NVIDIA Isaac Sim, OpenCV, PCL, SOLIDWORKS, Fusion360, Onshape, Linux (Ubuntu)
- Languages:** English; Japanese